

Indian Statistical Institute
End-semestral Examination: 2024-25
Programme: Master of Mathematics
Course: Number Theory

Maximum Marks: 60

Duration: 3 Hours

Date: May 5, 2025

Instruction: Answer all questions. The following results can be applied without proof.

$$\sum_{1 \leq n \leq x} \frac{1}{n} = \log x + \gamma + O(1/x), \text{ for } x > 1,$$

$$\sum_{2 \leq n \leq x} \log n = x \log x - x + O(\log x),$$

$$\sum_{1 \leq n \leq x} d(n) = x \log x + (2\gamma - 1)x + O(\sqrt{x}).$$

1. Suppose $\psi(X) := \sum_{n \leq X} \Lambda(n)$, where Λ denotes the von Mangoldt function. Show that

$$\psi(X) \geq cX$$

for some constant $c > 0$ for all X sufficiently large. [14]

2. For each of the sums below, find an asymptotic formula with an explicit main term and one or more error terms of lower order of magnitude:

$$-x \log x - (3/2 - 5/4)x + O(\sqrt{x}) \quad (a) \sum_{n \leq X} \left\{ \frac{X}{n} \right\}, \quad (b) \sum_{n \leq X} \left\{ \frac{X}{n} \right\} \log n, \quad -x(\log x)^2 - 3(2-1)x \log x + 3(2-1)x + O(\sqrt{x} \log x)$$

where $\{.\}$ denotes the fractional part. [7+7=14]

3. Consider $\mathbb{Q}_3$ with 3-adic topology. Show that (a) the sequence $(2^n)_{n \geq 1}$ is not convergent (Hint: produce two distinct limit points)

(b) the sequence $(4^{3^n})_{n \geq 1}$ is convergent and find the limit. [7+7=14]

4. Show that $L(1, \chi) \neq 0$, where χ is a Dirichlet character that takes non-real values. [14]

5. (a) Suppose the Legendre symbol $\left(\frac{-5}{p}\right) = 1$, where p is a prime number. Show that p is not a prime element in the ring of integers of $\mathbb{Q}(\sqrt{-5})$.

(b) Suppose $d > 0$ is square-free and $d \equiv 2$ or $3 \pmod{4}$. Assuming that there is a solution to $x^2 - dy^2 = 1$ with $y \neq 0$, show that in the interval $(1, \infty)$, there is a smallest unit in the ring of integers of $\mathbb{Q}(\sqrt{d})$. [7+7=14]

0, 3, 63, 255, 1023, 4095

$x(n+1) > x_n$
 $x/n > \frac{x}{n+1}$
 $\psi(p) - \psi(p_{n+1}) > 0$
 $x < y$
 $\psi(x) < \psi(y)$
 $\psi(x) - \psi(y) < 0$

$$a \leq x \leq a+1$$

$$\lceil x \rceil = a$$

$$\sqrt{a} \leq \sqrt{a+1}$$